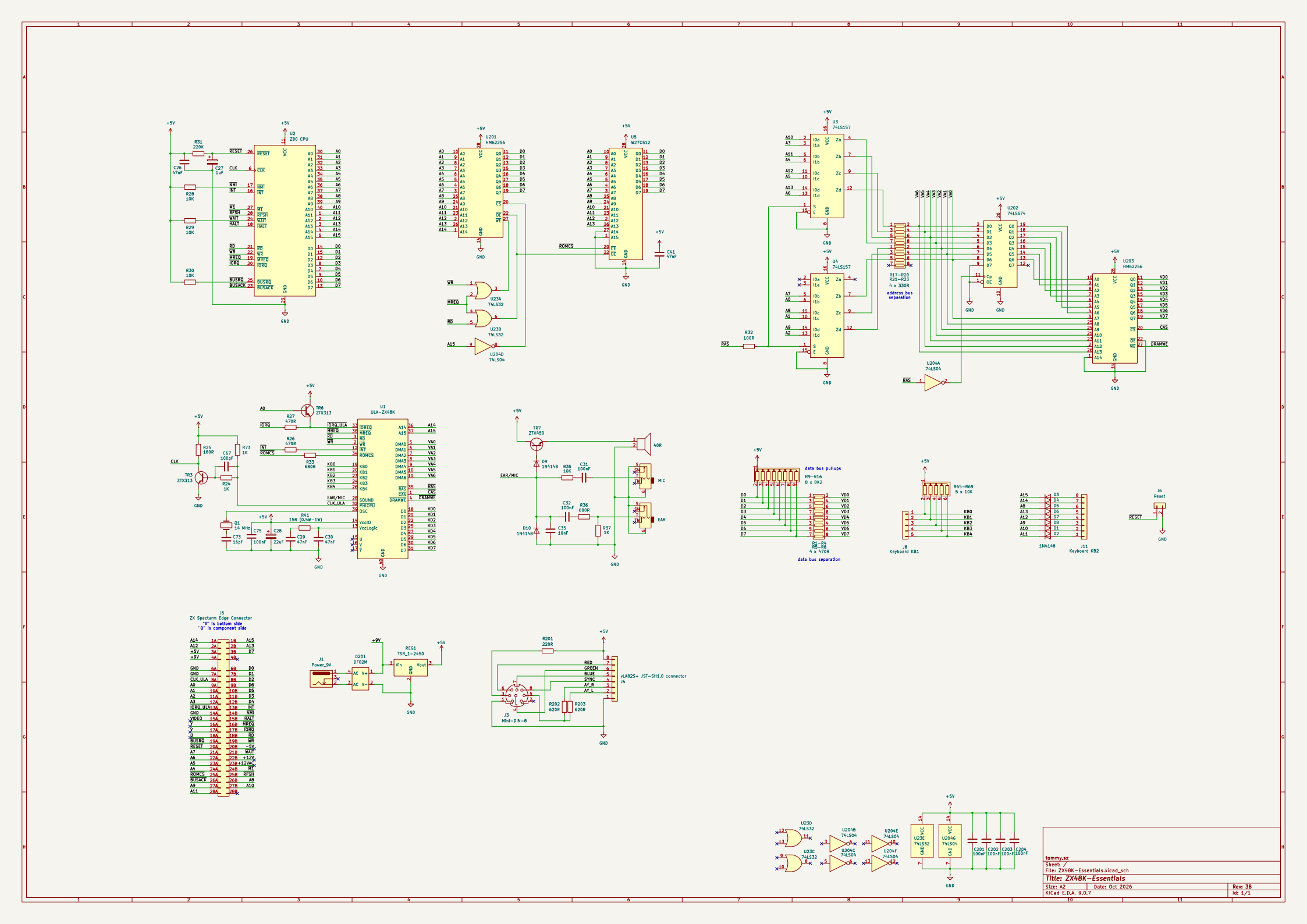

U2
Z80 CPU
U1
ULA-ZX48K
U201
HM62256
U5
W27C512
U3
74LS157
U4
74LS157
U202
74LS574
U203
HM62256
U23A
74LS32
U23B
74LS32
U204D
74LS04
U204A
74LS04
TR6
ZTX313
TR7
ZTX450
TR3
ZTX313
J5
ZX Spectrum Edge Connector
"A" is bottom side
"B" is component side
J1
Power_9V
D201
DF02M
REG1
TSR_1-2450
J3
Mini-DIN-8
J4
*LA82S+ JST-SH1.0 connector
J8
Keyboard KB1
J11
Keyboard KB2
J6
Reset
data bus pullups
R9–R16
8 x 8K2
data bus separation
R1–R4
R5–R8
4 x 470R
address bus
separation
R17–R20
R21–R23
4 x 330R
R65–R69
5 x 10K
U23D
74LS32
U23C
74LS32
U204B
74LS04
U204C
74LS04
U204E
74LS04
U204F
74LS04
U23E
74LS32
U204G
74LS04
tommy.sz
Sheet: /
File: ZX48K-Essentials.kicad_sch
Title: ZX48K-Essentials
Size: A2
Date: Oct 2026
Rev: 3B
KiCad E.D.A. 9.0.7
Id: 1/1